

**IM.Sc. (CS) Sem.-8 Examination
Web Data Management**

Time : 3.00 Hours]**July-2025****[Max.Marks : 70****Instructions:**

- Write both the Sections in the separate answer book.
- Both Sections having equal weightage.
- Draw Diagrams wherever necessary.
- Make Assumptions wherever necessary.

SECTION – I

Q-1 Do as Directed. [Any Seven] [7]

1. Define a well-formed XML document.
2. What is an XML namespace used for?
3. Differentiate between an XML attribute and an element.
4. What is the role of the **let** clause in XQuery?
5. What is the purpose of XLink in XML?
6. Name any two axes used in XPath.
7. Define ontology.
8. Define full-text indexing.

Q-2 Attempt the following:

a. Describe the XML Data Model with an example. How does XML represent data hierarchically? [7]

b. Differentiate between structured, semi-structured, and unstructured data with examples. How is XML used to represent semi-structured data? [7]

OR

Q-2 Attempt the following:

c. Explain the characteristics and challenges of modeling web data. Give examples of semi-structured data. [7]

d. What are the basic components of XML? Explain XML syntax rules with examples. [7]

Q-3 Attempt the following: [7]

a. What is XQuery? Explain FLWOR expressions in detail with an example. [7]

b. Compare DTD and XSD. [7]

OR [7]

Q-3 Attempt the following:

c. What is XPath? Explain steps and path expressions with examples. [7]

d. What is XQuery? Describe doc() and collection() with suitable example.

SECTION - II

- Q-4 Do as Directed. [Any Seven] [7]
1. What is the purpose of web analytics?
 2. Define page tagging.
 3. Name one open-source web analytics tool.
 4. What does Adwords help analyze?
 5. Give an example of a recommendation system.
 6. What does KPI stand for in web analytics?
 7. Define "XLink."
 8. Define "XPointer".
- Q-5 Attempt the following: [7]
- a. Describe the architecture of a web search engine. Explain parsing and indexing phases. [7]
 - b. Explain following terms with example in Distributed system.
1) Reliability 2) Scalability 3) Availability 4) Efficiency
- OR
- Q-5 Attempt the following: [7]
- c. Compare and contrast page tags and log files in web analytics. Which provides better insight? [7]
 - d. Describe the features, benefits, and limitations of Google Analytics.
- Q-6 Attempt the following: [7]
- a. Explain the role of recommendation systems in web-scale applications with an example. [7]
 - b. What are ontologies? Explain their role in semantic web with a real-world example.
- OR
- Q-6 Attempt the following: [7]
- c. What is distributed computing? Explain its use in building scalable web applications. [7]
 - d. Explain Web crawling process in details. [7]